

A Polynomial-Time Approximation Scheme for Constrained Assortment Optimization under the Mallows Model

Abstract

We study assortment optimization when customers' preferences follow a single Kendall–Mallows distribution over products and an outside option. The seller maximizes expected revenue subject to a cardinality constraint and one knapsack constraint, either separately or jointly. We develop a deterministic polynomial-time approximation scheme that returns a feasible assortment with revenue at least $(1 - \epsilon)$ times optimal in $N^{O(1/\epsilon)} \text{poly}(\mathcal{L}, 1/\epsilon)$ bit operations. The result allows an arbitrary position of the outside option and imposes no bounds on the dispersion parameter or the ratio of positive revenues. The analysis uses a positive integral representation of choice probabilities. Retaining the first $O(1/\epsilon)$ offered alternatives permits the remaining product factors to be compressed with uniform relative error. Rounded cumulative weights then yield a layered dynamic program, and profit scaling preserves the original resource budget exactly. Rational lower approximations of the integrals and exact evaluation of final candidates complete the bit-complexity argument. We also derive certified branch bounds and report finite computational checks of the construction.

Keywords: assortment optimization; Mallows model; approximation algorithms; dynamic programming; resource constraints.

1 Introduction

A seller choosing an assortment must account for the way customers substitute among the available products. Under a ranking model, a customer draws a preference order and selects the highest-ranked available alternative, including the option of making no purchase. This representation links an assortment decision directly to substitution behavior, but it also makes each product's demand depend on the other products offered. Cardinality and resource limits add a further requirement: a high-revenue assortment must fit the seller's available space or budget.

We consider preferences generated by a single Kendall–Mallows distribution. The model assigns probabilities to complete rankings according to their inversion distance from a central order. Its structure offers a potential route to tractable optimization, although choosing an assortment still involves searching over exponentially many subsets. The position of the outside option is consequential. When it can appear anywhere in the central order, products on either side of it may contribute to revenue. A method that optimizes only small assortments therefore needs an additional justification, and it cannot be assumed that an early segment of the central ranking contains the entire solution.

The question addressed here is whether the single-model problem admits an approximation scheme that is polynomial for each fixed accuracy while retaining the original cardinality and capacity restrictions. We allow arbitrary nonnegative rational revenues and resource requirements, and an arbitrary rational dispersion parameter. The approximation must remain valid when individual purchase probabilities are very small: their products with large revenues may still be important to the objective. These features favor relative, componentwise error control over a uniform additive approximation of choice probabilities.

Our main result is a deterministic polynomial-time approximation scheme (PTAS). For every $\epsilon \in (0, 1)$, it returns an assortment satisfying the original constraints and earning at least $(1 - \epsilon)$ OPT. Its bit complexity is $N^{O(1/\epsilon)} \text{poly}(\mathcal{L}, 1/\epsilon)$, where N is the number of alternatives including the outside option and $\mathcal{L}$ is the input encoding length. The running time has no

dependence on the numerical magnitude of the capacity beyond its encoding length. The result concerns a single preference distribution and one resource budget, with an optional cardinality restriction.

The central step is a compression of the choice probabilities. We express each probability as a positive integral and retain the factors associated with the first K offered alternatives in the central order. These retained factors control the relative error from replacing the remaining factors by an exponential. The bound is $O(1/K)$ uniformly over the number and scale of the remaining factors. Consequently, the identities of the later offered products affect the approximate objective through a total weight and a prefix weight. Rounding these two aggregates produces a polynomial number of states for each guessed head. The tail remains a decision variable and may contain many products.

Two implementation issues are part of the guarantee. First, resource use is kept exact while path profits are scaled, so the approximation does not require additional capacity. Second, the integral values are computed from below with a certified relative error in rational arithmetic. Final assortments are compared by their exact revenues. Together, these steps make the guarantee valid in the binary input model, including near the boundary values of the dispersion parameter. The resulting degree in N is large; the contribution is an approximation guarantee, and the computational checks reported here are intended to examine its ingredients rather than establish scalability.

Section 2 relates the result to the closest work. Section 3 specifies the model and main theorem. Section 4 develops the structural approximation, and Section 5 gives the algorithm and its analysis. Section 6 discusses branch bounds and computational checks. Section 7 concludes. Supporting proofs and reproducibility details appear in the appendices.

2 Related literature

The ranking distribution introduced by Mallows (1957) provides the probabilistic basis of our model. For assortment optimization, Désir et al. (2021) develop efficient choice-probability evaluation, a compact mixed-integer formulation, and approximation schemes for Mallows-based demand. Their results include a PTAS when the outside option is last in every central ranking; that approach also accommodates downward-closed feasibility constraints. This establishes a useful comparison: imposing a resource constraint alone does not distinguish the present result. Our theorem allows the outside option to have any position, while restricting demand to one Mallows component and the resource system to one knapsack and cardinality.

Rieger and Segev (2024) give a quasi-polynomial-time approximation scheme for the general single-Mallows assortment problem without structural or parametric restrictions. Their result is the closest approximation benchmark for the unconstrained special case studied here. Theorem 3.1 gives polynomial running time for fixed accuracy under rational input encoding and also enforces the stated resource restrictions. Its structural argument controls the error after retaining $O(1/\epsilon)$ offered alternatives. The tail is then optimized through aggregate weights, so the construction does not require a small near-optimal assortment. This comparison concerns the stated single-model setting; it does not extend the guarantee to mixtures.

The capacity subproblem is related to resource-constrained path approximation. Hassin (1992) develops fully polynomial approximation schemes for the restricted shortest-path problem. We use the same general principle of scaling objective values while retaining an exact resource label, and give a self-contained maximization argument for the acyclic graph arising here. The Mallows-specific content lies in obtaining that graph with polynomially many states and a uniform relative approximation of revenue.

3 Model and main result

There are N alternatives, indexed by their positions in a fixed central ranking $1, \dots, N$. Alternative $o \in [N]$ is the outside option. It is always offered and has revenue $r_o = 0$ and resource use $d_o = 0$. For $0 < q < 1$, a permutation σ of the *entire* universe has probability

$$\Pr(\sigma) = \frac{q^{\text{inv}(\sigma)}}{Z_N(q)}, \quad Z_n(q) = \prod_{j=1}^n (1 + q + \dots + q^{j-1}), \quad Z_0(q) = 1. \quad (3.1)$$

Here $\text{inv}(\sigma)$ is its inversion count relative to the central ranking. The customer chooses the first alternative in A encountered in σ . Writing this probability as $P_i(A)$, the revenue is

$$R(A) = \sum_{i \in A} r_i P_i(A), \quad o \in A.$$

The available constraints are

$$|A \setminus \{o\}| \leq C, \quad \sum_{i \in A} d_i \leq B. \quad (3.2)$$

Either constraint may be omitted. We assume $r_i, d_i, B \geq 0$, integer $C \geq 0$, and binary rational encoding for all numerical inputs, including q and the requested accuracy. Let $\mathcal{L}$ denote their total encoding length. The capacity is a static resource budget, such as shelf space. The optimization problem is to maximize $R(A)$ over these feasible assortments; its optimal value is denoted by OPT . The distribution and all product attributes are known to the seller when the assortment is chosen. For $q = 0$, the ranking is the central order deterministically; for $q = 1$, it is uniform over all permutations.

Theorem 3.1 (Polynomial-time approximation scheme). *For rational $q \in [0, 1]$ and $0 < \epsilon < 1$, there is a deterministic algorithm returning an assortment $\hat{A}$ that contains o , satisfies every imposed constraint in (3.2), and obeys*

$$R(\hat{A}) \geq (1 - \epsilon) \text{OPT}.$$

Its bit complexity is $N^{O(1/\epsilon)} \text{poly}(\mathcal{L}, 1/\epsilon)$. No lower bound on q , $1 - q$, or positive choice probabilities, and no upper bound on the ratio of positive revenues, is assumed.

The model retains the original rank gaps between offered alternatives. Replacing it by a Mallows distribution on the offered set would change the problem. Mixtures of central rankings, multiple independent capacity resources, and dynamic inventories are outside the theorem's scope.

Feasibility in Theorem 3.1 means that the returned assortment obeys the original budget B and product limit C , with no multiplicative or additive relaxation. The approximation concerns expected revenue alone. As usual, the theorem is a PTAS statement: the exponent of N depends on $1/\epsilon$. It does not assert a fully polynomial scheme or a running time of the form $f(1/\epsilon) \text{poly}(N, \mathcal{L})$.

The rational encoding of q is also substantive. If the input instead specifies a rational θ and sets $q = e^{-\theta}$, exact rational q is not thereby available. Extending the bit-complexity statement to that input convention would require a certified real-arithmetic model or a perturbation analysis. Throughout the paper, q itself is the rational input.

4 Structural approximation of assortment revenue

The approximation has two stages. We first replace the factors associated with later offered alternatives by an exponential while retaining a short head. We then round only the aggregate weights in that exponential. Both operations produce pointwise lower bounds on revenue with relative losses that depend on the retained head size.

4.1 A positive representation of choice probabilities

For $0 < q < 1$, set

$$w_i = q^{i-1}, \quad c_q = \frac{1-q}{-\log q}, \quad D_n(t) = \prod_{j=0}^n (1 + q^j t). \quad (4.1)$$

The following representation separates the original rank weights from the assortment-dependent factors. Its positivity will permit relative error control for each revenue contribution.

Proposition 4.1 (Positive choice integral). *For every nonempty $A \subseteq [N]$ and $i \in A$,*

$$P_i(A) = c_q w_i \int_0^\infty \prod_{\substack{j \in A \\ j \leq i}} (1 + w_j t)^{-1} \prod_{\substack{j \in A \\ j \geq i}} (1 + q w_j t)^{-1} dt. \quad (4.2)$$

In particular, i contributes two factors, with coefficients w_i and $q w_i$.

The proof is given in Appendix A.

For $q = 1$, all full rankings are equally likely, so $P_i(A) = 1/|A|$. Formula (4.2) continues to hold with $c_q = w_i = 1$, since its integrand then equals $(1+t)^{-|A|-1}$. All subsequent arguments apply at $q = 1$ with these definitions. The case $q = 0$ is treated in Section 5.6.

The weights w_i retain the positions in the original central order. Thus the representation applies directly to nonconsecutive offer sets. Because the integrand is nonnegative, a common relative lower bound on the integrals remains a relative lower bound after multiplication by revenues and summation.

4.2 Uniform relative error from compression

The next lemma explains why retaining a short head is sufficient. The retained factors concentrate the normalized integrand near the origin, where the product-to-exponential approximation has a controlled error. The bound depends on the number of retained factors and is uniform over the number and size range of the factors being compressed.

Lemma 4.2 (Product-to-exponential compression). *Let $K > 3$ be an integer. Let $C(t) = \prod_\ell (1 + a_\ell t)^{-1}$ have at least K coefficients $a_\ell \geq a > 0$; any additional coefficients are nonnegative. Let $0 \leq b_j \leq a$ and $B_0 = \sum_j b_j$. Define*

$$I = \int_0^\infty C(t) \prod_j (1 + b_j t)^{-1} dt, \quad J = \int_0^\infty C(t) e^{-B_0 t} dt.$$

Then, with $\alpha_K = 1/[4(K-3)]$,

$$(1 - \alpha_K)I \leq J \leq I. \quad (4.3)$$

Proof. Let f denote the integrand of I , and put $z = B_0/a$. Since $x/(1+xt)$ increases in x and $b_j \leq a$,

$$-\frac{f'(t)}{f(t)} \geq \frac{Ka}{1+at} + \sum_j \frac{b_j}{1+at} = \frac{a(K+z)}{1+at}.$$

Thus $f(t)/(1+at)^{-(K+z)}$ is nonincreasing. After normalization, this monotone likelihood ratio implies that f/I is stochastically no larger than the normalized density proportional to $(1+at)^{-(K+z)}$: the two densities have at most one crossing, from above to below. In particular,

$$\mathbb{E}_f[t^2] \leq \frac{2}{a^2(K+z-2)(K+z-3)}. \quad (4.4)$$

Here and below the subscript f means normalization by its integral. For $x \geq 0$, $0 \leq x - \log(1+x) \leq x^2/2$. Therefore

$$0 \leq D(t) := B_0 t - \sum_j \log(1 + b_j t) \leq \frac{t^2}{2} \sum_j b_j^2 \leq \frac{a B_0 t^2}{2}.$$

Since $J/I = \mathbb{E}_f[e^{-D(t)}]$,

$$0 \leq 1 - J/I \leq \frac{a B_0}{2} \mathbb{E}_f[t^2] \leq \frac{z}{(K+z-2)(K+z-3)} \leq \frac{1}{4(K-3)}.$$

The last inequality follows from $z/(c+z)^2 \leq 1/(4c)$ for $c = K-3 > 0$. $\square$

Lemma 4.3 (Additive shift). *Suppose C has at least $K > 2$ factors with coefficients at least $a_0 > 0$, and any additional nonnegative factors. For $L(s) = \int_0^\infty e^{-st} C(t) dt$ and $s, d \geq 0$,*

$$\left(1 - \frac{d}{a_0(K-2)}\right) L(s) \leq L(s+d) \leq L(s). \quad (4.5)$$

The proof is given in Appendix B.

4.3 Retaining the first offered alternatives

We now apply Lemma 4.2 to the assortment problem. Fix

$$K = \lceil 1/\epsilon \rceil + 4. \quad (4.6)$$

Enumerate every feasible assortment with fewer than K alternatives separately and evaluate its true revenue exactly using Appendix A.3. Every other assortment has a unique head H consisting of its first K offered alternatives. Write $h = \max H$ and $A = H \cup T$, where $T \subseteq \{h+1, \dots, N\}$. We enumerate all possible K -element heads, discard heads violating a resource bound, and enforce

$$o \leq h \implies o \in H, \quad o > h \implies o \in T. \quad (4.7)$$

No unselected position at or before h is later inserted into T .

For a fixed head define

$$L_H(s) = \int_0^\infty e^{-st} \prod_{j \in H} (1 + w_j t)^{-1} dt, \quad (4.8)$$

$$L_{H,i}(s) = \int_0^\infty e^{-st} \prod_{\substack{j \in H \\ j \leq i}} (1 + w_j t)^{-1} \prod_{\substack{j \in H \\ j \geq i}} (1 + q w_j t)^{-1} dt, \quad i \in H. \quad (4.9)$$

Let

$$W(T) = \sum_{j \in T} w_j, \quad U_i(T) = \sum_{\substack{j \in T \\ j < i}} w_j.$$

The compressed revenue is

$$\tilde{R}_H(T) = c_q \left[\sum_{i \in H} r_i w_i L_{H,i}(qW(T)) + \sum_{i \in T} r_i w_i L_H(qW(T) + (1-q)U_i(T) + w_i) \right]. \quad (4.10)$$

Proposition 4.4 (Pointwise compressed approximation). *For every tail T ,*

$$(1 - \alpha_K) R(H \cup T) \leq \tilde{R}_H(T) \leq R(H \cup T). \quad (4.11)$$

The proof is given in Appendix B.

For a fixed head, the approximate revenue of a tail product depends on the tail only through $W(T)$ and $U_i(T)$. Conditioning on the total $W(T)$ will therefore leave a sequential decision problem whose state is a prefix weight. The next step makes both aggregates discrete without introducing a dependence on the ratio of the largest and smallest rank weights.

4.4 Discretizing aggregate weights

Put $a = w_h$ and use the rational grid

$$\Delta = \frac{a}{4N}, \quad z_i = \left\lceil \frac{w_i}{\Delta} \right\rceil, \quad \bar{w}_i = \Delta z_i \quad (i > h), \quad M = 4N^2. \quad (4.12)$$

Because $0 < w_i \leq a$, we have $1 \leq z_i \leq 4N$ and $\sum_{i \in T} z_i \leq M$. Replace only W and U_i in (4.10) by $\bar{W} = \sum_{j \in T} \bar{w}_j$ and $\bar{U}_i = \sum_{j \in T, j < i} \bar{w}_j$. Keep the multipliers $r_i w_i$ and the standalone w_i in each exponent unchanged. Call the result $\bar{R}_H(T)$.

Proposition 4.5 (Weight rounding). *With $\beta_K = 1/[4(K-2)]$,*

$$(1 - \beta_K) \tilde{R}_H(T) \leq \bar{R}_H(T) \leq \tilde{R}_H(T). \quad (4.13)$$

The proof is given in Appendix B.

5 The approximation algorithm

For each candidate head, the rounded approximation can be optimized on a layered graph. Cardinality is represented by a state coordinate, and a knapsack constraint is enforced through minimum-resource labels. We describe these components before combining their approximation factors.

5.1 A layered graph for a fixed head and total

Use one-sided rational integral values $(1 - \delta)L(s) \leq \underline{L}(s) \leq L(s)$ with $\delta = \epsilon/4$; Section 5.3 supplies their construction and bit complexity. Write $\underline{E}_H(T)$ for the resulting rounded, compressed score with the common factor c_q omitted. Fix a head H and enumerate $Q \in \{0, \dots, M\}$. It represents the exact integer sum $\sum_{i \in T} z_i$. All tails considered in this branch must end with this total. The head score is the nonnegative rational constant

$$b_H(Q) = \sum_{i \in H} r_i w_i \underline{L}_{H,i}(q\Delta Q). \quad (5.1)$$

Scan positions $i = h + 1, \dots, N$ in order. At prefix weight p , selecting i has nonnegative rational score

$$b_i(p; Q) = r_i w_i \underline{L}_H(q\Delta Q + (1 - q)\Delta p + w_i). \quad (5.2)$$

The selection edge sends p to $p + z_i$; the skip edge keeps p and has zero score. Omit edges with $p + z_i > Q$. If $i = o$, its skip edge is prohibited, even though its revenue, count consumption, and resource consumption are zero. The source has $p = 0$ and accepted terminal states have $p = Q$. Unreachable states can be discarded. Each path selects a unique tail and has score $\underline{E}_H(T) - b_H(Q)$.

With a cardinality bound, let $C_H = C - |H \setminus \{o\}|$ and add a counter $k \in \{0, \dots, \min(C_H, N)\}$ to the state. Selecting a product increases it by one; selecting o leaves it unchanged. Skip transitions also preserve it. All terminal states with $p = Q$ are allowed. A zero-score, zero-resource edge to a common sink can be added to each. If there is no cardinality bound, omit this state coordinate.

Proposition 5.1 (No physical capacity constraint). *For a fixed H, Q , a longest-path dynamic program on this graph exactly maximizes the rational lower score, including when a cardinality constraint is imposed.*

Proof. The state records all information on which future edge scores and the remaining cardinality allowance depend. The graph is acyclic because every item transition advances a layer. The standard recurrence $V(v) = \max_{e=(u,v)} \{V(u) + b_e\}$, initialized by $V(\text{source}) = 0$ and $-\infty$ elsewhere, therefore enumerates all feasible tails implicitly. Add (5.1) to the terminal value. $\square$

5.2 Enforcing a resource budget

Suppose $\sum_{i \in A} d_i \leq B$ is imposed. For a fixed head, the remaining budget is $B_H = B - \sum_{i \in H} d_i$. Give a selection edge for i resource use d_i and give skip and sink edges resource use zero. The graph already incorporates the cardinality state when that constraint is present. Each path has at most N edges: there are at most $N - K$ item layers and at most one additional sink edge.

Lemma 5.2 (Resource-constrained path FPTAS). *Let a finite acyclic source–sink graph have nonnegative rational edge profits b_e and resource uses d_e , with at most N edges on any source–sink path. For any rational budget $B_H \geq 0$ and $0 < \eta < 1$, a deterministic algorithm finds a feasible path of profit at least $(1 - \eta)G^*$, or determines that no feasible path exists. It performs $O(|E|N^2/\eta)$ label updates, in addition to acyclic shortest-path computations, with polynomial rational bit overhead. Here G^* is the maximum feasible profit.*

The proof is given in Appendix C.

Apply this lemma with $\eta = \epsilon/4$. Adding the fixed head contribution preserves the factor: for $b_H(Q) \geq 0$,

$$b_H(Q) + (1 - \eta)G^* \geq (1 - \eta)(b_H(Q) + G^*).$$

The returned path respects the original rational budget. In particular, the label dimension depends on N and $1/\eta$, not on the numerical magnitude of B_H . Capacity is never rounded or relaxed.

The proof first removes edges that belong to no feasible path. The largest remaining edge profit is then a lower bound on the optimal path profit, and at most N times this value is an upper bound. This scale makes the number of profit labels polynomial. For a fixed label, keeping the smallest exact resource use preserves every feasible continuation.

5.3 Rational scores and exact final comparisons

The path algorithms require rational profits with a controlled relative error. An additive quadrature tolerance would not suffice uniformly for small choice probabilities multiplied by large revenues. The following lemma supplies the required one-sided oracle in the binary input model.

Lemma 5.3 (Rational relative lower approximation). *Let $m \geq 2$, $c_1, \dots, c_m > 0$, $s \geq 0$, and $0 < \delta < 1$ be rational. For*

$$L = \int_0^\infty f(t) dt, \quad f(t) = e^{-st} \prod_{\ell=1}^m (1 + c_\ell t)^{-1}, \quad (5.3)$$

one can compute a rational $\underline{L}$ satisfying

$$(1 - \delta)L \leq \underline{L} \leq L \quad (5.4)$$

in time polynomial in the input encoding length and $1/\delta$.

The proof is given in Appendix D.

Use this oracle with $\delta = \epsilon/4$ on every nonzero-revenue integral in a large-set branch. Zero multipliers are kept exactly zero. Define $F(A) = R(A)/c_q$. Let $\underline{F}_H(T)$ be the unscaled rounded score with every integral replaced by its rational lower approximation. Then

$$\rho F(H \cup T) \leq \underline{F}_H(T) \leq F(H \cup T), \quad \rho = (1 - \delta)(1 - \beta_K)(1 - \alpha_K). \quad (5.5)$$

The common positive factor c_q is omitted from comparisons *within* each branch. Small assortments and all returned branch candidates are instead compared by their exact rational true revenues

from Appendix A.3. No comparison mixes a scaled surrogate with a true revenue, and no logarithm is needed for candidate selection.

The oracle uses a geometric mesh on a truncated interval. Monotonicity bounds the integrated discretization error, while rational lower approximations of the exponential control evaluation error. Algebraic and exponential tail bounds are combined to determine the truncation endpoint. These operations depend polynomially on encoding length even when the integral is very small.

For final candidate evaluation, the exact formula in Appendix A.3 uses $O(N^3)$ rational operations per assortment, with polynomial bit lengths. The integral approximation is therefore needed to construct branch scores; the comparison among returned assortments is exact.

5.4 Algorithm and approximation guarantee

The preceding components give Algorithm 1. Every large assortment belongs to exactly one head and rounded-total branch, and every returned branch candidate is feasible. The proof tracks the losses from compression, weight rounding, integration, and path optimization.

Algorithm 1 (Assortment PTAS for $0 < q \leq 1$).

1. Set $K = \lceil 1/\epsilon \rceil + 4$, $\delta = \epsilon/4$, and $\eta = \epsilon/4$. Include the feasible assortment $\{o\}$ as an initial candidate with true revenue zero.
2. Enumerate all feasible $A \ni o$ with $|A| < K$. Compute their true revenues exactly using Appendix A.3.
3. Enumerate all K -element heads H . Discard those violating (4.7) or a head resource bound. Form (4.12) and, for $Q = 0, \dots, M$, construct the graph in Section 5.1.
4. Solve each feasible branch by exact longest path when capacity is absent, or by Lemma 5.2 when it is present. For the returned tail T , evaluate $R(H \cup T)$ exactly and update the best feasible candidate.
5. Return the candidate with largest exact true revenue.

If $K > N$, Step 2 covers all assortments and Step 3 is empty. This is consistent with the stated PTAS time bound.

Proof of Theorem 3.1 for $0 < q \leq 1$. Let A^* be optimal. If $|A^*| < K$, it is evaluated exactly and the algorithm returns an optimal assortment. Otherwise let H^* be its unique first K offered alternatives, $T^* = A^* \setminus H^*$, and $Q^* = \sum_{i \in T^*} z_i$ for the grid of H^* . This branch is enumerated, and T^* is a feasible path in it. By (5.5) and the branch optimization guarantee, the tail $\hat{T}$ returned by this branch satisfies

$$F(H^* \cup \hat{T}) \geq \underline{F}_{H^*}(\hat{T}) \geq (1 - \eta)\underline{F}_{H^*}(T^*) \geq (1 - \eta)\rho F(A^*).$$

The final exact comparison yields $R(\hat{A}) \geq R(H^* \cup \hat{T})$. Multiplication by $c_q > 0$ therefore gives

$$\begin{aligned} R(\hat{A}) &\geq (1 - \eta)(1 - \delta)(1 - \beta_K)(1 - \alpha_K) \text{OPT} \\ &\geq (1 - \epsilon/4)^4 \text{OPT} \geq (1 - \epsilon) \text{OPT}. \end{aligned}$$

Here $\alpha_K, \beta_K \leq \epsilon/4$ and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $0 \leq x_j \leq 1$. Without capacity the factor $(1 - \eta)$ can be replaced by one. The constructed paths and the small-set enumeration enforce every constraint. The argument also covers $\text{OPT} = 0$ without division by the optimal revenue. $\square$

Table 1: Upper bounds on state and label updates, where $K = \lceil 1/\epsilon \rceil + 4$.

Constraints	Total updates across heads and totals
None	$O(N^{K+5})$
Cardinality	$O(N^{K+6})$
One capacity	$O(N^{K+7}/\epsilon)$
Cardinality and one capacity	$O(N^{K+8}/\epsilon)$

5.5 Running time

There are at most N^K heads. For a fixed head and total Q , there are $O(NM)$ states and edges without a count coordinate, and $O(N^2M)$ with it. There are $M + 1 = O(N^2)$ choices of Q . Table 1 gives bounds on label or edge updates; oracle evaluation and rational arithmetic are additional polynomial factors: The separate small-set enumeration is no larger in asymptotic order. Each returned candidate is evaluated using $O(N^3)$ rational operations by Appendix A.3. There are at most $O(N^{K+2})$ large-set candidates, so exact candidate evaluation fits within the displayed overall bounds. For a head branch $K \leq N$. Each $w_i = q^{i-1}$ has bit length $O(N\mathcal{L})$; the grid, shifts, and exact multipliers also have polynomial length. Lemma 5.3 produces polynomial-length rational profits. Finite sums of at most N such profits or costs, comparisons of labels, and profit floors preserve polynomial bit complexity. Combining these observations proves the stated $N^{O(1/\epsilon)}$ $\text{poly}(\mathcal{L}, 1/\epsilon)$ bound. Integral values can be computed once per graph edge and reused across all profit labels. Only reachable rounded totals need be enumerated; a subset-sum DP over $\{0, \dots, M\}$, enforcing outside inclusion and optionally cardinality, finds these totals in polynomial time.

5.6 Boundary values of the dispersion parameter

For $q = 0$, the full ranking is the central ranking deterministically. If a positive-revenue product is chosen, it must precede o . Removing every other product preserves that choice and feasibility because costs and count uses are nonnegative. Thus scan all $i < o$ for which $\{i, o\}$ is feasible, return the highest-revenue singleton product together with o , or return $\{o\}$ if none improves revenue. This is an exact algorithm and completes the theorem, including $N = 1$, $C = 0$, $B = 0$, and zero resource costs.

At $q = 1$ and without a physical capacity bound, there is also a simple exact algorithm. For each permitted product count k , the best assortment consists of the k highest-revenue products together with o , since its revenue is their revenue sum divided by $k + 1$. Check $0 \leq k \leq \min\{C, N - 1\}$, interpreting $C = N - 1$ when cardinality is unrestricted. With a capacity bound, the general PTAS remains applicable.

6 Implementation and computational verification

6.1 Branch bounds and the choice of head size

The algorithm's largest combinatorial cost is the enumeration of heads. Once a head and rounded total have been fixed, a relaxed path problem provides a certified bound that can eliminate unpromising branches. Let $U_{H,Q}$ be the maximum rational lower score on the branch after dropping physical capacity, while retaining cardinality and outside-option restrictions and including $b_H(Q)$. Equation (5.5) implies $R(H \cup T) \leq c_q U_{H,Q}/\rho$ for every feasible completion. The relative-error guarantee thus converts a maximized lower surrogate into an upper bound on the true branch optimum.

Table 2: Complete-algorithm checks on eight alternatives. Returned revenues and optimal revenues coincide in these six instances.

q	Outside rank o	C	B	$ \hat{A} $	$R(\hat{A}) = \text{OPT}$	Branches
1	1	–	–	8	7/8	28
1	5	6	–	7	6/7	27
1	8	–	6	7	6/7	28
1	5	6	6	7	6/7	27
1/2	1	–	6	7	63/127	28
1/2	8	6	6	7	10752/10795	28

All pruning comparisons can be made rationally. A certified rational upper bound $c_q^+ \geq c_q$ gives the safe bound $c_q^+ U_{H,Q}/\rho$; in particular, $c_q^+ = 1$ is always valid. A branch may be discarded when this bound does not exceed the exact revenue of the current feasible incumbent. Lagrangian penalties for capacity give additional bounds. Appendix E derives these expressions, a tighter computable c_q^+ , and the guarantee for an arbitrary prescribed head size $K \geq 4$. These bounds preserve the theorem, but no reduction in worst-case enumeration is asserted.

The prescribed head size can be substantial: $\epsilon = 0.1$ gives $K = 14$. Using a smaller head reduces enumeration and yields the explicit approximation factor in Appendix E, with a corresponding loss in the guarantee. This tradeoff motivates implementation choices, but its computational value must be assessed on instances larger and more varied than those below.

6.2 Design of the computational checks

The computations examine distinct parts of the construction using independent finite references. Complete permutation enumeration supplies exact choice probabilities. Enumeration of feasible paths checks the dynamic programs, and enumeration of feasible assortments supplies true optimal revenues. The probability comparisons use rational arithmetic. The numerical integral comparisons use 50 decimal digits and a relative tolerance of 10^{-35} .

The complete-algorithm examples have $N = 8$ alternatives, unit product revenues and unit resource uses, and zero revenue and resource use for the outside option. We set $\epsilon = 3/4$, so $K = 6$, and vary the outside position, dispersion, and constraint combination as shown in Table 2. An omitted constraint is denoted by a dash. These small examples permit exhaustive comparison and include solutions from the large-assortment branches. Their deliberately loose accuracy parameter makes those branches observable at this instance size.

In these six cases, branch profits are obtained from high-precision quadrature and converted to 35-digit rational values. Resource comparisons, profit labels, and final true-revenue comparisons use exact rational arithmetic. The examples therefore test the complete combinatorial construction, but they are not full implementations of the certified integral oracle on every edge. The certified oracle is exercised separately on eight integrals and on every score in two additional branch-bound examples.

6.3 Results and interpretation

All six complete-algorithm examples attain their exactly enumerated optimum. The checks of 3,076 choice probabilities and 49 symbolic identities also agree exactly with their references. Across 204 numerical integral identities, the largest relative error is below $6 \cdot 10^{-51}$. The separate path tests include a constructed profit-rounding tie that returns 19/10 against an optimum of 2. Its relative loss of 1/20 confirms that profit scaling can lose revenue even when the approximation guarantee holds.

For the certified branch-bound checks, we use a head of six alternatives and two tail products, with $q \in \{1, 99/100\}$. Five nonnegative rational Lagrangian multipliers produce ten bounds, all verified against exact branch optima using rational arithmetic. The relaxed-capacity and penalized bounds are not necessarily tight. For example, when $q = 1$, the branch optimum is $8/7$, the zero-multiplier bound is approximately 1.84610, and the bound at $\lambda = 1/2$ is approximately 1.64343. This example checks validity and illustrates possible tightening; it does not measure overall pruning speed.

These results support consistency of the formulas and implementations on the specified finite cases. The approximation theorem follows from the analysis in Sections 4–5. Neither the unit-revenue examples nor the component checks establish practical performance for large, heterogeneous instances.

7 Conclusion

We have developed a PTAS for single-Mallows assortment optimization with an arbitrarily positioned outside option, cardinality, and one resource budget. The structural result is a uniform relative approximation after the first $O(1/\epsilon)$ offered alternatives are retained. It permits the remaining assortment to be optimized through rounded total and prefix weights, even when individual rank weights and revenues span large ranges. Exact resource labels and rational lower integration preserve the original constraints and make the algorithm valid in the binary input model.

The analysis also identifies the main limitation of the construction. Its running time is dominated by head enumeration, and the present computational checks do not resolve whether the certified branch bounds make that search useful at practical scales. A stronger algorithm would need to reduce this enumeration cost while preserving the uniform relative-error bound. Extending the approach to multiple preference components or multiple resource budgets would additionally require a replacement for the single-order aggregate state or the scalar minimum-resource label used here; neither extension follows from the current argument.

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A Choice probabilities and exact revenue evaluation

A.1 Geometric-product moments

The probability representation rests on the following moment identity. For $0 < q < 1$, use the notation in (4.1).

Lemma A.1 (Geometric-product moments). *For $n \geq 1$ and $0 \leq k \leq n-1$,*

$$c_q \int_0^\infty \frac{t^k}{D_n(t)} dt = q^{-k(k+1)/2} \frac{Z_k(q) Z_{n-k-1}(q)}{Z_n(q)}. \quad (\text{A.1})$$

Proof. Write $F_{n,k}$ for the integral on the left before multiplication by c_q . For $g(t) = D_{n-1}(t)^{-1}$,

$$\frac{g(qt) - g(t)}{t} = \frac{1 - q^n}{D_n(t)}.$$

Integrating on $[a, b]$ and changing variables in the first term gives

$$\int_a^b \frac{g(qt) - g(t)}{t} dt = \int_{qa}^a \frac{g(u)}{u} du - \int_{qb}^b \frac{g(u)}{u} du.$$

Since $g(0) = 1$ and $g(\infty) = 0$, the limit as $a \downarrow 0$, $b \uparrow \infty$ is $-\log q$. Thus $F_{n,0} = (-\log q)/(1 - q^n)$. For $1 \leq k \leq n-1$, both of the following integrals converge:

$$F_{n,k-1} + q^n F_{n,k} = F_{n-1,k-1}, \quad F_{n,k-1} + F_{n,k} = q^{-k} F_{n-1,k-1}.$$

Their difference yields

$$F_{n,k} = \frac{q^{-k} - 1}{1 - q^n} F_{n-1,k-1}.$$

Starting from $F_{n-k,0}$ and substituting $Z_j = (1 - q)^{-j} \prod_{\ell=1}^j (1 - q^\ell)$ proves (A.1). $\square$

A.2 Proof of the positive integral representation

Proof of Proposition 4.1. Let $U = [N] \setminus A$, and fix the set $B \subseteq U$ of alternatives appearing before i in a full ranking in which i wins. Put $k = |B|$. The ranking consists of the elements of B , then i , then its remaining elements. Summing the weights of the two internal permutations gives $Z_k Z_{N-k-1}$. The inversions between the three parts number

$$\begin{aligned} E(B, i) &= \sum_{b \in B} (b-1) - \frac{k(k-1)}{2} + (i-1) - |\{b \in B : b < i\}| \\ &= i-1 + \sum_{b \in B} (b-1 + \mathbf{1}_{\{b > i\}}) - \frac{k(k+1)}{2}. \end{aligned}$$

Consequently,

$$P_i(A) = \frac{w_i}{Z_N} \sum_{B \subseteq U} q^{-k(k+1)/2} Z_k Z_{N-k-1} \prod_{b \in B} q^{b-1 + \mathbf{1}_{\{b > i\}}}. \quad (\text{A.2})$$

For the full offer set, the denominator in (4.2) is $D_N(t)$. Removing an unoffered $b < i$ cancels $1 + q^{b-1}t$, and removing $b > i$ cancels $1 + q^b t$. Hence its integrand is

$$\frac{\prod_{b \in U} (1 + q^{b-1 + \mathbf{1}_{\{b > i\}}} t)}{D_N(t)}.$$

Expand the numerator and apply Lemma A.1 term by term. The result is exactly (A.2). $\square$

A.3 Exact rational revenue evaluation

For $0 < q < 1$, let $(q; q)_j = \prod_{\ell=1}^j (1 - q^\ell)$ with $(q; q)_0 = 1$. For a fixed offered $i \in A$, expand

$$\prod_{\substack{j \notin A \\ j < i}} (1 + q^{j-1} z) \prod_{\substack{j \notin A \\ j > i}} (1 + q^j z) = \sum_{k=0}^{N-|A|} e_k^{(i)} z^k. \quad (\text{A.3})$$

Proposition 4.1 and Lemma A.1 imply the exact formula

$$\begin{aligned} P_i(A) &= \frac{w_i}{Z_N} \sum_{k=0}^{N-|A|} e_k^{(i)} q^{-k(k+1)/2} Z_k Z_{N-k-1} \\ &= (1-q) w_i \sum_{k=0}^{N-|A|} e_k^{(i)} q^{-k(k+1)/2} \frac{(q; q)_k (q; q)_{N-k-1}}{(q; q)_N}. \end{aligned} \quad (\text{A.4})$$

In contrast to its integral representation, this expression is rational for rational q and contains no logarithm.

The coefficients in (A.3) can be computed by a product-of-linear-factors recurrence. Initialize $e_0 = 1$ and all other coefficients to zero. For each factor $1 + vz$, update in descending degree order by

$$e_k \leftarrow e_k + v e_{k-1}.$$

There are at most N factors and at most $N + 1$ coefficients, so this requires $O(N^2)$ rational operations for one offered alternative. Precompute Z_j and the required powers of q ; summing (A.4) costs $O(N)$ more operations. Repeating for every $i \in A$ and summing $r_i P_i(A)$ computes $R(A)$ in $O(N^3)$ rational operations.

The bit lengths are polynomial as well. If $q = a/b$ is in lowest terms, all exponents in these products and coefficients have magnitude $O(N^2)$. The elementary symmetric coefficients are sums of at most 2^N monomials, and the partition functions have degree $O(N^2)$ and total coefficient mass at most $N!$. Common-denominator representations and products of polynomially many such rational quantities therefore have polynomial bit length in $N, \mathcal{L}$. Exact signs and comparisons of candidate revenues are thus polynomial-time operations. At $q = 1$, use $P_i(A) = 1/|A|$; at $q = 0$, use the deterministic rule from Section 5.6.

Let $\mathcal{C}$ be the candidates generated by the branch algorithms together with the small assortments. Exact final selection satisfies

$$R(\arg \max_{A \in \mathcal{C}} R(A)) \geq R(A') \quad \text{for every } A' \in \mathcal{C}.$$

In particular it weakly dominates selecting the same candidates by their lower scores, while preserving the overall PTAS complexity.

B Supporting structural proofs

B.1 Additive shifts

Proof of Lemma 4.3. The logarithmic derivative of $e^{-st} C(t)$ is at most $-K a_0 / (1 + a_0 t)$. The likelihood-ratio argument from the proof of Lemma 4.2, now with reference density proportional to $(1 + a_0 t)^{-K}$, gives $\mathbb{E}_s[t] \leq 1/[a_0(K - 2)]$. Use $L(s + d)/L(s) = \mathbb{E}_s[e^{-dt}] \geq 1 - d\mathbb{E}_s[t]$. Monotonicity gives the upper bound. $\square$

B.2 Head and tail compression

Proof of Proposition 4.4. For $i \in H$, retain the $K + 1$ factors in (4.9). Their coefficients are at least qw_h , whereas all removed factors have coefficients $qw_j \leq qw_h$. Their sum is $qW(T)$, so Lemma 4.2 applies. For $i \in T$, retain the K head factors $w_j \geq w_h$. All remaining coefficients are at most w_h , and their sum equals

$$\sum_{\substack{j \in T \\ j < i}} w_j + w_i + q \sum_{\substack{j \in T \\ j \geq i}} w_j = qW(T) + (1 - q)U_i(T) + w_i.$$

Apply the same lemma to each choice integral, then sum using $r_i w_i \geq 0$. $\square$

B.3 Rounding aggregate weights

Proof of Proposition 4.5. Every rounded sum increases by at most $N\Delta = a/4$. For a head term its exponent increases by at most $qa/4$; its core has at least K coefficients at least qa . Lemma 4.3 gives loss at most $1/[4(K - 2)]$. For a tail term the shift is at most $q(a/4) + (1 - q)(a/4) = a/4$, and its core has K coefficients at least a . The same bound applies. All coefficients multiplying the integrals are nonnegative. This also proves the upper bound by monotonicity. $\square$

C Resource-constrained path approximation

Proof of Lemma 5.2. Compute the minimum resource use $\ell_s(u)$ from the source to each vertex, and $\ell_t(v)$ from each vertex to the sink. If the minimum full-path resource exceeds B_H , the branch is infeasible. Otherwise an edge $e = (u, v)$ occurs on some feasible full path exactly when

$$\ell_s(u) + d_e + \ell_t(v) \leq B_H. \quad (\text{C.1})$$

Necessity is immediate. For sufficiency concatenate minimum-resource prefix and suffix paths with e . They cannot repeat a vertex in an acyclic graph. Delete all other edges. Every originally feasible path remains; the witness path for each retained edge remains too, since each of its edges belongs to that feasible path.

Let $P = \max_e b_e$ among retained edges. Nonnegative profits and a feasible path containing an edge of profit P imply

$$P \leq G^* \leq NP. \quad (\text{C.2})$$

If $P = 0$, return any feasible path. Otherwise set $\tau = \eta P/N$ and $\hat{b}_e = \lfloor b_e/\tau \rfloor$. Each $\hat{b}_e \leq N/\eta$, so a path's rounded profit is at most $\lfloor N^2/\eta \rfloor$. For every vertex v and rounded profit z , maintain

$$D(v, z) = \text{minimum original rational resource use to reach } v \text{ with profit label } z.$$

Initialize $D(\text{source}, 0) = 0$ and all other entries to $+\infty$. In topological order apply

$$D(v, z + \hat{b}_e) \leftarrow \min\{D(v, z + \hat{b}_e), D(u, z) + d_e\}.$$

Retaining the smallest resource use for a label loses no feasible continuation. Choose the largest terminal label whose stored resource is at most B_H . If π^* is an optimal feasible path and $\hat{\pi}$ is returned, then

$$\begin{aligned} G(\hat{\pi}) &\geq \tau \sum_{e \in \hat{\pi}} \hat{b}_e \geq \tau \sum_{e \in \pi^*} \hat{b}_e \geq G^* - N\tau \\ &= G^* - \eta P \geq (1 - \eta)G^*. \end{aligned}$$

Feasibility follows from exact resource sums and comparisons. There are $O(N^2/\eta)$ labels per vertex and the stated number of updates. All profit and resource inputs are rational with polynomial bit length, as are their path sums and the profit-scaling operations. $\square$

D Rational evaluation of the integrals

Proof of Lemma 5.3. Relabel the coefficients so that $c_1 \geq c_2 \geq \dots \geq c_m > 0$; their order does not affect the integral. Let $S = s + \sum_{\ell} c_{\ell}$. Since $1 + x \leq e^x$, $f(t) \geq e^{-St}$ and $L \geq 1/S$. Also $f(t) \leq 1$ and $f(t) \leq (c_1 c_2 t^2)^{-1}$ for $t > 0$. Set

$$\ell = \frac{\delta}{8S}, \quad U_{\text{alg}} = \frac{8S}{\delta c_1 c_2}, \quad \gamma = 1 + \frac{\delta}{8}. \quad (\text{D.1})$$

For $s > 0$, choose the least nonnegative integer κ such that $2^{\kappa} \geq 8S/(\delta s)$ and put

$$U_{\text{exp}} = \frac{\kappa}{s}, \quad U = \min\{U_{\text{alg}}, U_{\text{exp}}\}.$$

For $s = 0$, put $U = U_{\text{alg}}$. All cutoffs are rational, and κ can be found by integer comparisons without evaluating a logarithm. The integrals over $[0, \ell]$ and $[U_{\text{alg}}, \infty)$ are each at most $\delta/(8S)$. When $s > 0$, the exponential tail gives

$$\int_{U_{\text{exp}}}^{\infty} f(t) dt \leq e^{-\kappa}/s \leq 2^{-\kappa}/s \leq \delta/(8S).$$

Thus the smaller cutoff U has the same tail guarantee. Moreover $U > \ell$: $S^2 \geq 4c_1 c_2$, $S \geq s$, and $\kappa \geq 1$. Form the rational geometric mesh $x_0 = \ell$, $x_{j+1} = \min\{\gamma x_j, U\}$ until $x_J = U$. Because f decreases, its right-endpoint sum Q is a lower bound on L . For every $t \in [x_j, x_{j+1}]$, $x_{j+1} \leq \gamma t$, so

$$\begin{aligned} Q &= \sum_{j=0}^{J-1} (x_{j+1} - x_j) f(x_{j+1}) \\ &\geq \int_{\ell}^U f(\gamma t) dt = \gamma^{-1} \int_{\gamma \ell}^{\gamma U} f(u) du \\ &\geq \gamma^{-1} L - \ell - \frac{\delta}{8\gamma S} \geq \left(1 - \frac{3\delta}{8}\right) L. \end{aligned}$$

This integrated comparison is essential: no pointwise relative control of $e^{-s\gamma t}/e^{-st}$ is needed.

Set $\nu = \delta/(16SU)$. At a node y , compute a rational E_y with $0 \leq e^{-sy} - E_y \leq \nu$. Divide it by the exact rational product $\prod_{\ell}(1 + c_{\ell}y)$, then round down to a dyadic grid of mesh $2^{-b} \leq \nu$. The resulting $\underline{f}(y)$ lies in $[0, f(y)]$ and has absolute error at most 2ν . The sum using $\underline{f}$ instead of f loses at most $2\nu U = \delta/(8S) \leq \delta L/8$. Thus

$$(1 - \delta/2)L \leq \underline{L} := \sum_j (x_{j+1} - x_j) \underline{f}(x_{j+1}) \leq L,$$

which implies the asserted, slightly weaker guarantee.

For completeness, the exponential lower approximation also uses rational arithmetic only. Choose an integer D with $2^{-D} \leq \nu$. If $x = sy \geq D$, take $E_y = 0$, since $e^{-x} \leq 2^{-D}$. Otherwise $0 \leq x < D$. Let $P_n = \sum_{j=0}^n x^j/j!$ and $v_{n+1} = x^{n+1}/(n+1)!$. Choose n so that $x/(n+2) \leq 1/2$ and $2v_{n+1} \leq \nu$. The positive Taylor tail is at most $2v_{n+1}$, hence

$$P_n \leq e^x \leq P_n + 2v_{n+1}, \quad E_y = (P_n + 2v_{n+1})^{-1} \leq e^{-x}, \quad e^{-x} - E_y \leq 2v_{n+1} \leq \nu.$$

The last estimate uses $P_n \geq 1$. A factorial bound shows that $n = O(D + \log(1/\nu))$ suffices.

If the rational input has total bit length b_0 , then $\log(U/\ell)$ and $\log(1/\nu)$ are polynomial in $b_0 + \log(1/\delta)$. The number of cells is $O(\delta^{-1} \log(U/\ell) + 1)$; mesh points, Taylor terms, dyadic approximations, and the final rational sum all have polynomial bit length. Even multiplication of all denominators gives a polynomial bound, since there are polynomially many terms of polynomial length. All comparisons, floors, products, and sums can therefore be performed in polynomial bit time. $\square$

E Certified branch bounds and accuracy tradeoffs

The head enumeration remains the main worst-case bottleneck. The following optional bounds permit safe pruning while preserving the approximation theorem. They also make explicit the scaling needed to compare a branch surrogate with an exact true-revenue incumbent.

E.1 Capacity relaxation and Lagrangian bounds

Fix H, Q and keep all count and outside-option restrictions in the graph. Let $U_{H,Q}$ be the maximum unscaled rational lower score, including $b_H(Q)$, over all source–sink paths after relaxing physical capacity. From (5.5), every feasible completion satisfies

$$R(H \cup T) \leq \frac{c_q}{\rho} F_H(T) \leq \frac{c_q}{\rho} U_{H,Q}. \quad (\text{E.1})$$

Thus maximizing a pointwise lower surrogate and dividing by its guaranteed relative factor gives an upper bound on true branch revenue.

When capacity is imposed, every rational $\lambda \geq 0$ yields the alternative unscaled bound

$$U_{H,Q}(\lambda) = b_H(Q) + \lambda B_H + \max_{\pi} \sum_{e \in \pi} (b_e - \lambda d_e), \quad (\text{E.2})$$

where the maximization drops physical capacity but retains the graph’s other restrictions. For a feasible path, adding $\lambda(B_H - \sum_e d_e) \geq 0$ to its score proves that (E.2) bounds its unscaled lower score from above. Signed edge rewards cause no difficulty for a longest-path DP on a DAG. Taking the minimum over any finite set of nonnegative rational multipliers, including zero, preserves validity and cannot weaken the zero-multiplier bound. No claim that a positive multiplier always improves it is required.

E.2 A rational comparison with true revenue

The constant c_q is generally irrational. For a safe rational upper bound, let

$$I_0 = \int_0^\infty \frac{dt}{(1+t)(1+qt)} = \frac{1}{c_q}.$$

This follows directly from Proposition 4.1 for $A = \{1\}$. For any rational $0 < \delta_c < 1$, evaluate it once with Lemma 5.3, obtaining $(1 - \delta_c)I_0 \leq \underline{I}_0 \leq I_0$. Then

$$\frac{1 - \delta_c}{\underline{I}_0} \leq c_q \leq c_q^+ := \min \left\{ 1, \frac{1}{\underline{I}_0} \right\}. \quad (\text{E.3})$$

The additional bound $c_q \leq 1$ follows from $-\log q \geq 1 - q$. At $q = 1$, simply use $c_q^+ = 1$. Both entries in the minimum are valid upper bounds, so the minimum is safe.

Consequently

$$\text{UB}_{H,Q} = \frac{c_q^+}{\rho} U_{H,Q}, \quad \text{UB}_{H,Q}(\lambda) = \frac{c_q^+}{\rho} U_{H,Q}(\lambda) \quad (\text{E.4})$$

are rational upper bounds on the true feasible branch optimum. A branch can be discarded if one such bound is at most the exact true revenue of an already feasible incumbent. If the optimal branch is discarded, the incumbent already matches or exceeds its best revenue, so pruning preserves Theorem 3.1. The accuracy δ_c affects bound tightness only and consumes none of the revenue-approximation budget. One may also use the simpler, weaker upper bound $c_q^+ = 1$ without the extra oracle call. The head contribution must be included in every bound, and the factor c_q must not be silently dropped when comparing with a true-revenue incumbent.

E.3 Explicit accuracy for a chosen head size

For any implementation choice $K \geq 4$, numerical relative tolerance $\delta_{\text{num}} \in (0, 1)$ and path tolerance $\delta_{\text{path}} \in [0, 1)$, the same argument yields the explicit factor

$$\left(1 - \frac{1}{4(K-3)}\right) \left(1 - \frac{1}{4(K-2)}\right) (1 - \delta_{\text{num}})(1 - \delta_{\text{path}}).$$

Here $\delta_{\text{path}} = 0$ when the branch is solved by exact longest path. All smaller assortments are evaluated exactly. This formula states an accuracy–runtime tradeoff; the prescribed K in (4.6) gives the PTAS. The complete algorithm is not claimed to be an EPTAS or an FPTAS.